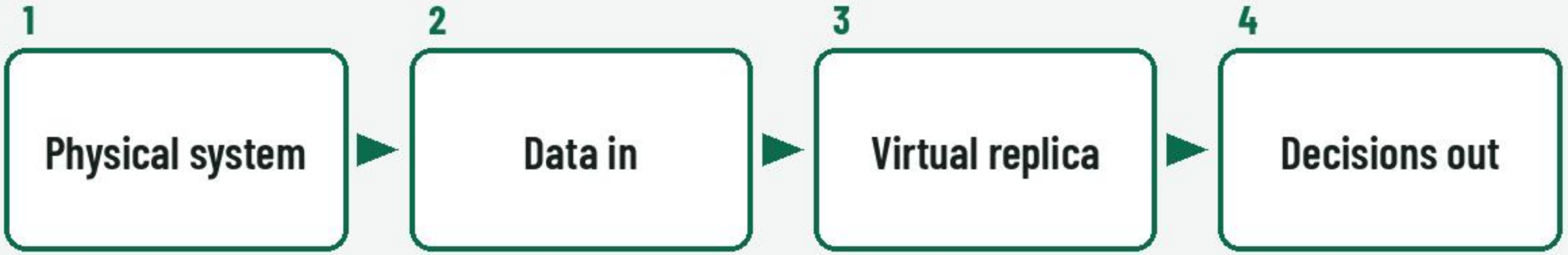
FOUNDATIONS

# Digital twin concepts

Module 1 · What a twin is, and why transport needs one

Transport Twin Study Path

# Three parts and a loop



# Model, shadow or twin?

## Data in

Model: manual

Shadow: automatic

Twin: automatic

## Decisions out

Model: manual

Shadow: manual

Twin: automatic

# Five questions a twin answers

- What is happening now?
- Why is it happening?
- What happens next?
- What if we try X?
- What should we do, and do it?

# Why transport is a hard case

- Millions of people making choices
- Assets that last 50 to 100 years
- Many owners for one trip
- Decisions from seconds to decades

# Start with the decision

## Decision → clock → data

Pick one decision, find its time scale, then choose sensors and models